Day 3 Problem Set

Geometry Summer Credit | Transformations, Rotations, and Dilations

Name:	Date:	Period:

Goal

Practice coordinate rules for translations, reflections, rotations, and dilations. Use prime notation for image points.

Support Practice: Transformation Rules

1.	Translate (2, -5) by <3, 7>.

2.	Reflect (-4, 9) across the x-axis.

3.	Reflect (-4, 9) across the y-axis.

4.	Rotate (5, -1) 180 degrees about the origin.

5.	Dilate (-6, 10) by scale factor 1/2 from the origin.

Core Practice: Translations and Reflections

6.	Triangle A(-1, 3), B(4, 3), C(2, 6) is translated by <-5, -2>. Find A', B', and C'.

7.	Quadrilateral J(-3, -1), K(1, -1), L(2, 2), M(-2, 2) is reflected across the x-axis. Find the image coordinates.

8.	Reflect triangle P(2, -4), Q(6, -4), R(4, -1) across the y-axis.

9.	A point is translated from A(-7, 4) to A'(2, -1). What translation vector was used?

10.	Error analysis: A student translates (3, -2) by <-4, 5> and gets (7, -7). Explain and correct the error.

Day 3 Problem Set

Continued

Core Practice: Rotations and Dilations

11.	Rotate (-2, 8) 90 degrees counterclockwise about the origin.

12.	Rotate (9, -3) 180 degrees about the origin.

13.	Dilate (5, -7) by scale factor 3 from the origin.

14.	Dilate (-12, 6) by scale factor 1/3 from the origin.

15.	Triangle A(2, 2), B(6, 2), C(4, 8) is rotated 90 degrees counterclockwise about the origin.

16.	Triangle D(-1, 3), E(2, 3), F(1, 6) is dilated by scale factor 2 from the origin.

17.	A dilation maps (4, -8) to (1, -2). What is the scale factor?

18.	Error analysis: A student rotates (4, 7) 90 degrees counterclockwise and gets (7, 4). Explain and correct the error.

Mixed Review and Reasoning

19.	Spiral: Find the slope through (-2, 5) and (4, -1).

20.	Spiral: Same-side interior angles are labeled $3x + 12$ and $5x - 8$. Find x.

21.	Challenge: Describe a sequence of two transformations that maps $(2, 3)$ to $(-7, 4)$.

Before You Submit

Circle one rule you know well and star one problem where a diagram helped you.